

Parkinson's disease

- Parkinson's disease is a neurodegenerative condition which affects the basal ganglia and which presents with differing combinations of slowness of movement (bradykinesia), increased tone (rigidity), tremor and loss of postural reflexes.
- Prevalence rates are similar throughout the world, though lower rates have been reported for China and West Africa.

Age : Whilst 10% of the patients are under 45 years at presentation, the incidence and prevalence both increase with age, the latter rising to over 1 % in those over 60.

Sex : Sex incidence is about equal.

Cigarette Smoke : It is less common in cigarette smokers.

Progression of disease

- If symptoms start in middle life, the disease is usually steadily progressive and likely to shorten lifespan because of the complications of immobility and tendency to fall.
- Onset after 70 is unlikely to shorten life or become severe.

Aetiology and Pathophysiology

- A small number of cases are familial in nature and mutations in several genes have now been identified as an underlying cause.
- However, in the majority the cause is unknown, and no strong genetic factors have been identified.
- The discovery that methyl-phenyl-tetrahydropyridine (MPTP) caused severe parkinsonism in young drug users suggests that the idiopathic disease might be due to an environmental toxin
- There are several features, including depletion of the pigmented dopaminergic neurons in the

substantia nigra, hyaline inclusions in nigral cells (Lewy bodies), atrophic changes in the substantia nigra and depletion of neurons in the locus coeruleus.

- Reduced dopaminergic output from the substantia nigra to the globus pallidus leads to reduced inhibitory effects on the subthalamic nucleus, neurons of which become more active than usual in inhibiting activation of the cortex. This in turn results in bradykinesia.

Clinical features

- The classical syndrome of tremor, rigidity and bradykinesia may be absent initially.
- The non-specific symptoms of tiredness, aching limbs, mental slowness, depression and small handwriting (micrographia) may be noticed.
- The presentation is almost always unilateral, a resting tremor in an upper limb being a common presenting feature.
- The tremor may eventually affect the legs, mouth and tongue.
- It may remain the predominant symptom for some years.
- Bradykinesia may develop gradually.
- Most patients have difficulty with rapid fine movements, and this manifests itself as slowness of gait and difficulty with tasks such as fastening buttons, shaving or writing.
- Rigidity, or increased muscular tone, causes stiffness and a flexed posture.
- Postural righting reflexes are impaired early on in the disease, but falls tend not to occur until later.
- As the disease advances, speech becomes softer and indistinct.
- Although the features are initially unilateral, gradual bilateral involvement is the rule.
- Muscle strength and reflexes remain normal,

and plantar responses are flexor.

- There is a paucity of facial expression (hypomimia) and the blink reflex may be exaggerated and fail to habituate (glabellar tap sign).
- Eye movements are normal to standard clinical testing, provided allowance is made for the normal limitation of upward gaze with age.
- Sensation is normal and intellectual faculties are not affected initially. As the disease progresses, about one-third of patients develop cognitive impairment.

Physical abnormalities in Parkinson's disease

General

- Expressionless face
- Greasy skin
- Soft, rapid, indistinct speech
- Flexed posture
- Impaired postural reflexes

Gait

- Slow to start walking
- Shortened stride
- Rapid, small stride length, tendency to shorten (destination)
- Reduced arm swing
- Impaired balance on turning

Tremor

Resting (4-6 Hz) :

- Coarse, complex movements, usually first in fingers/thumb Flexion/extension of fingers

Abduction/adduction of thumb

Supination/pronation of forearm :

- May affect arms, legs, feet, jaw, tongue
- Intermittent, present at rest and when distracted
- Diminished on action

Postural (8-10 Hz) :

- Less obvious, faster, finer amplitude
- Present on action or posture, persists with movement

Rigidity :

- Cogwheel type, mostly upper limbs
- Plastic (lead pipe) type, mostly legs

Bradykinesia :

- Slowness in initiating or repeating movements
- Impaired fine movements, especially of fingers

Diagnosis and Investigations :

- The diagnosis is made clinically, as there is no diagnostic test for Parkinson's disease.
- Sometimes it is necessary to investigate patients to exclude other causes of Parkinsonism if there are any unusual features.

Tests for Wilson's diseases and imaging (CT or MRI)

- Patients presenting before the age of 50 are usually tested for Wilson's diseases, and imaging (CT or MRI) of the head may be needed if there are any features suggestive of pyramidal, cerebellar or autonomic involvement, or the diagnosis is otherwise in doubt.

Stroke

1. Acute stroke is characterised by the rapid appearance (usually over minutes) of a focal deficit of brain function, most commonly a hemiplegia with or without signs of focal higher cerebral dysfunction (such as aphasia), hemisensory loss, visual field defect or brain-stem deficit.
2. Provided that there is a clear history of a rapid-onset focal deficit, the chance of the brain lesion being anything other than vascular is 5%, or less.
3. However, care needs to be taken to exclude other differential diagnoses if the symptoms progress over hours or days.
4. Confusion, memory or balance disturbance are more often due to causes other than stroke.

Several terms have been used to classify strokes, often based on the duration and evolution of symptoms

Transient Ischaemic Attack (TIA) :

1. Describes a stroke in which symptoms resolve within 24 hours-all arbitrary cutoff which has little value in practice apart from perhaps indicating that underlying cerebral

haemorrhage or extensive cerebral infarction is extremely unlikely

2. The term TIA traditionally also includes patients with transient monocular blindness (also known as amaurosis fugax), usually due to a vascular occlusion in the retina.

Clinical features

Transient symptoms, such as syncope, amnesia, confusion and dizziness, which do not reflect focal cerebral dysfunction, are often mistakenly attributed to TIA

Stroke

1. This is the term reserved for those events in which symptoms last more than 24 hours.
2. The differential diagnosis of patients with symptoms lasting a few minutes or hours is similar to those with persisting symptoms.

Progressing stroke (or stroke in evolution)

1. the focal neurological deficit worsens after the patient first presents
2. Such worsening may be due to increasing volume of infarction, haemorrhagic transformation or increasing oedema

Completed stroke

1. The focal deficit persists and is not progressing.

Aetiology

Stroke risk factors

Fixed

- Age
- Gender (male > female, except in the very young and very old)
- Race (Afro-Caribbean > Asian > European)
- Heredity
- Previous vascular event, e.g. myocardial infarction, stroke or peripheral embolism
- High fibrinogen

Modifiable

- High blood pressure

- Heart disease (atrial fibrillation, heart failure, endocarditis)
- Diabetes mellitus
- Hyperlipidaemia
- Smoking
- Excess alcohol consumption
- Polycythaemia
- Oral contraceptives
- Social deprivation

Pathophysiology

Of patients presenting with a stroke



85% will have sustained a cerebral infarction due to inadequate blood flow to part of the brain



The remaining .i.e. 15% will have had an intra cerebral haemorrhage



Brain imaging is required to distinguish these pathologies



To guide management



The combination of severe headache and vomiting at the onset of the focal neurological deficits increases the likelihood of a haemorrhagic stroke.



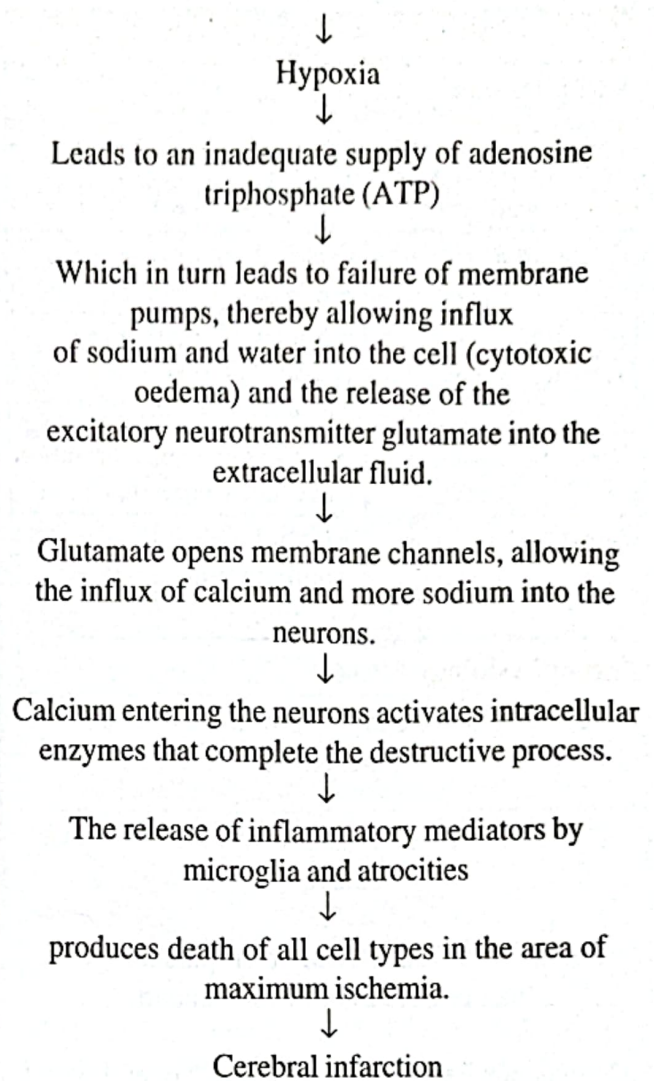
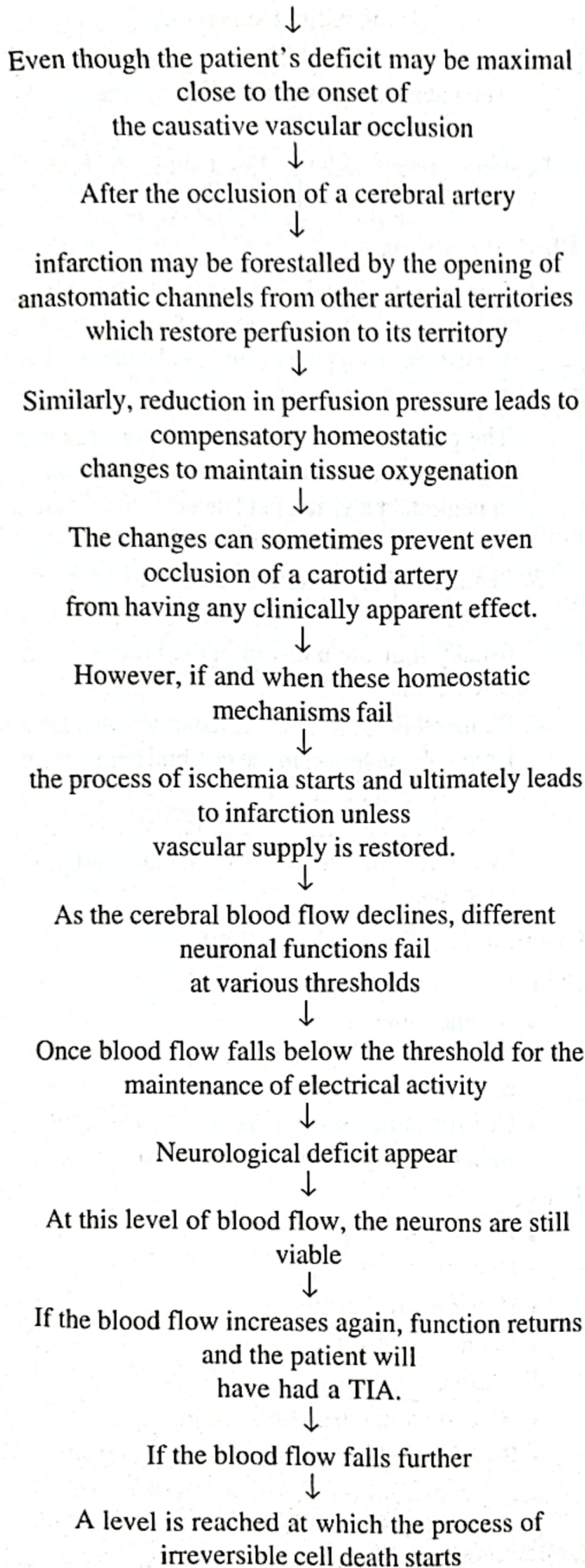
Cerebral infarction

Aetiology

1. Cerebral infarction is mostly due to thromboembolic disease secondary to atherosclerosis in the major extra cranial arteries (carotid artery and aortic arch)
2. About 20% of infarctions are due to embolism from the heart
3. A further 20% are due to intrinsic disease of small perforating vessels (lenticulostriate arteries), producing so called 'lacunar' infarctions.

Pathophysiology

Cerebral infarction is a process which takes some hours to complete



Intracerebral haemorrhage

1. This usually results from rupture of a blood vessel within the brain parenchyma
2. It may also occur in patient with a subarachnoid haemorrhage if the artery ruptures, into the brain substance as well as into the subarachnoid space.
3. Haemorrhage frequently occurs into an area of brain infarction.
4. If the volume of haemorrhage is large, this may be difficult to distinguish from primary intracerebral haemorrhage both clinically and radiologically.

The risk factors and underlying causes of intracerebral haemorrhage are as follows : Causes of intracerebral haemorrhage and associated risk factors :

Disease	Risk factors
Complex small vessel disease	Age
With disruption of vessel wall	Hypertension Low cholesterol
Amyloid angiopathy	Familial (rare) Age
Impaired blood clotting	Anticoagulant therapy Blood dyscrasia Thrombolytic therapy
Vascular anomaly	Arteriovenous malformation Cavernous haemangioma
Substance misuse	Alcohol Amphetamines Cocaine

Pathophysiology

The explosive entry of blood into the brain parenchyma
 ↓
 Causes immediate cessation of function in that area
 ↓
 As neurons are structurally disrupted and white matter fibre tracts are split apart.
 ↓
 The haemorrhage itself may expand over the first minutes or hours,
 ↓
 Or it may be associated with a rim of cerebral oedema,
 ↓
 Which, along with the haematoma,
 ↓
 Acts like a mass lesion to cause progression of the neurological deficit.
 ↓
 Intracerebral haemorrhage
 ↓
 If the enough, this can cause shift of the intracranial contents,
 ↓
 producing transtentorial coning and sometimes rapid death
 ↓

If the patient survives,

↓
 The haematoma is gradually absorbed,

↓
 Leaving a haemosiderin-lined slit in the brain parenchyma

Clinical features

1. The neurological deficit can be identified from the patient's history and, if the deficit is persistent, from the neurological examination.
2. Features of cerebral hemisphere
 The presence of a unilateral motor deficit, a higher cerebral function deficit such as aphasia or neglect, or a visual field defect usually places the lesion in the cerebral hemisphere.
3. Features of brain stem or cerebellum
 Ataxia, diplopia and/or bilateral weakness usually indicate a lesion in the brain stem or cerebellum.
4. Reduced conscious level usually indicates a large volume lesion in the cerebral hemisphere but may result from a lesion in the brain stem or complications such as, obstructive hydrocephalus, hypoxia or severe systemic infection.

Examination of a stroke patient

Skin :

- Xanthelasma
- Rash (arteritis, splinter haemorrhages, livedo reticularis)
- Colour and temperature change (limb ischemia/deep venous thrombosis)

Eyes :

- Arcus senilis
- Diabetic changes
- Hypertensive changes
- Retinal emboli

Cardiovascular system :

- Heart rhythm (atrial fibrillation)
- Blood pressure (hypertension, hypotension)
- Jugular venous pressure (heart failure, hypovolaemia)

- Murmurs (sources of embolism)
- Peripheral pulses and bruits (generalised arteriopathy)

Respiratory system :

- Signs of pulmonary oedema
- Signs of respiratory infection

Abdomen :

- Palpable bladder (urinary retention)

Locomotor :

- Injuries sustained during collapse with stroke
- Co-morbidities which influence functional ability

Investigations

Investigation of a patient presenting with an acute stroke aims to confirm the vascular nature of the lesion, distinguish cerebral infarction from haemorrhage and identify the underlying vascular disease and risk factors

Investigation of a patient with an acute stroke

Diagnostic question	Investigation
Is it a vascular lesion?	CT/MRI
Is it ischemic or haemorrhagic?	CT/MRI
Is it a subarachnoid haemorrhage?	CT/lumbar puncture
Is there any cardiac source of embolism?	Electrocardiogram (ECG) 24-hour ECG Echocardiogram
What is the underlying vascular disease?	Duplex ultrasound of carotids Magnetic resonance angiography (MRA) CT angiography (CTA) Contrast angiography
What are the risk factors?	Full blood count Cholesterol Blood glucose
Is there an unusual cause?	ESR Serum protein electrophoresis Clotting/thrombophilia screen

Causes and investigation of acute stroke in young patients.

Cause	Investigation
Cerebral infarct	
Cardiac embolism	Echocardiography (including transoesophageal)
Premature atherosclerosis	Serum lipids
Arterial dissection	MRI Angiography
Thrombophilia	Protein C, protein S Antithrombin III Factor V Leiden, prothrombin
Homocystinuria	Urinary amino acids Methionine loading test
Antiphospholipid antibody syndrome	Anticardiolipin antibodies/lupus anticoagulant
SLE	Antinuclear antibodies
Vasculitis	ESR CRP Antineutrophil cytoplasmic antibody (ANCA)
CADASIL (cerebral autosomal dominant arteriopathy with subcortical infarcts and leucoencephalopathy)	MRI brain Genetic analysis Skin biopsy
Mitochondrial cytopathy	Serum lactate White cell mitochondrial DNA Muscle biopsy Mitochondrial molecular genetics
Fabry's disease	Alpha-galactosidase levels in blood
Neurovascular syphilis Primary intracerebral haemorrhage	Syphilis serology
Arteriovenous malformation (AVM)	MRI/angiography
Drug misuse	Drug screen (amphetamine, cocaine)
Coagulopathy	Prothrombin time (PT) and

	activated partial thromboplastin time (APTT) Platelet count
Subarachnoid haemorrhage	
Saccular ('berry') aneurysm	MRI/angiography
AVM	MRI/angiography
Vertebral dissection	MRI/angiography

Bell's palsy

Aetiology

1. Idiopathic facial nerve palsy or Bell's palsy is a common condition affecting all ages and both sexes.
2. The lesion is within the facial canal and may be due to reactivation of latent herpes simplex virus 1 infection.

Clinical features

1. Symptoms usually develop sub acutely over a few hours
2. Pain around the ear preceding the unilateral facial weakness.
3. Patients often describe the face as 'numb',
4. There is no objective sensory loss (except possibly to taste).
5. Hyperacusis can occur if the nerve to stapedius is involved
6. Diminished salivation and tear secretion.
7. Examination reveals an ipsilateral lower motor neuron facial nerve palsy.
8. Vesicles in the ear or on the palate indicate that the facial palsy is due to herpes zoster rather than Bell's palsy.
9. Prednisolone 40-60 mg daily for a week speeds recovery if started within 72 hours.
10. Artificial tears and ointment prevent exposure keratitis and the eye should be taped shut overnight.
11. About 80% of patients recover spontaneously within 12 weeks.
12. A slow or poor recovery is predicted by

complete paralysis, older age and reduced facial motor action potential amplitude after the first week.

13. Recurrence can occur but should prompt further investigation.
14. Aberrant re-innervations may occur during recovery, producing unwanted facial movements such as eye closure when the mouth is moved, or 'crocodile tears' (tearing during salivation).

Motor Neuron Disorders

Motor neuron disorders (MNDs) are a clinically and pathologically heterogeneous group of neurologic diseases characterized by progressive degeneration of motor neurons; they include both sporadic and hereditary diseases.

Either or both of the following two sets of motor neurons can be affected:

- Upper motor neurons (UMNs), which originate from the primary motor cortex of the cerebrum (precentral gyrus) and possess long axons forming corticospinal and corticobulbar tracts
- Lower motor neurons (LMNs), which originate in the brainstem (cranial nerve [CN] motor nuclei) and spinal cord (anterior horn cells) and directly innervate skeletal muscles
- MNDs can be classified into those affecting primarily the UMNs, those affecting primarily the LMNs, and those affecting both, and the nomenclature is used accordingly.
- The patient's symptoms vary, depending on which set of motor neurons is involved.

The common Motor neuron disorders

- Amyotrophic lateral sclerosis (ALS)
- Primary lateral sclerosis (PLS)
- Hereditary spastic paraparesis (HSP)
- Progressive bulbar palsy (PBP), including hereditary forms
- Spinal muscular atrophy (SMA)
- X-linked spinobulbar muscular atrophy (SBMA; Kennedy disease)

- Postpolio syndrome (PPS)

Amyotrophic lateral sclerosis (ALS)

- ALS, also known as Lou Gehrig disease, is the most common neurodegenerative disease of adult onset involving the motor neuron system.
- It is a fatal disorder and is characterized by progressive skeletal muscle weakness and wasting or atrophy (ie, amyotrophy), spasticity, and fasciculations as a result of degeneration of the UMNs and LMNs, culminating in respiratory paralysis
- The etiology of sporadic amyotrophic lateral sclerosis (ALS) remains uncertain, however, ALS is considered to be a multifactorial disease that is triggered by a complex interaction of internal factors (eg, genetic susceptibility to different neuronal insults and immunologic alterations) and external factors (eg, environmental variables).
- The key theories proposed to date for the pathogenesis of ALS include glutamate-induced excitotoxicity, oxidative injury, altered mitochondrial function, cytoskeleton alterations, axonal transport dysregulation, neuroinflammation, immunomodulation, and autoimmunity.

Primary lateral sclerosis

- The etiology of primary lateral sclerosis (PLS) is unknown, but it may be similar to that proposed for ALS.
- The age of onset for primary lateral sclerosis (PLS) is usually between 40 and 60 years, with spasticity in the legs accompanied by hyperactive deep tendon reflexes, clonus, and Babinski sign.
- Patients with spastic dysarthria as an initial symptom of PLS are also described. In contrast to ALS, the LMNs remain intact, and thus, no amyotrophy is observed.

Hereditary spastic paraparesis

- More than 40 genetic loci ascribed to the different types of hereditary spastic paraparesis (HSP) have been identified; however, a clear

genetic basis for most HSP types remains uncertain.

- The essential clinical findings are slowly progressive and often include severe spasticity, hyperreflexia, and weakness in a pyramidal distribution, noticeably in both lower limbs, with extensor plantar responses.

Progressive bulbar palsy

- The etiopathogenesis of Brown-Vialetto-Van Laere syndrome and Fazio-Londe disease remains elusive.
- Brown-Vialetto-Van Laere syndrome typically presents with progressive sensorineural hearing loss.

Spinal muscular atrophy

- Type I-III spinal muscular atrophy (SMA) is caused by alterations in the survival motor neuron gene (SMN).
- The clinical picture of spinal muscular atrophy (SMA) is highly variable and represents a continuum. The age of onset ranges from before birth to adulthood.

X-linked spinobulbar muscular atrophy (Kennedy disease)

- X-linked spinobulbar atrophy (SBMA) is clinically characterized by adult-onset limb and bulbar weakness, muscular atrophy, and fasciculation, with frequent occurrence of endocrine disturbances such as gynecomastia, testicular atrophy, hypercholesterolemia, and diabetes mellitus

Postpolio syndrome

- Most patients with postpolio syndrome (PPS) present with new, slowly progressive muscle weakness, frequently accompanied by muscle pain (myalgias) and fatigue, which can occur in both previously affected and unaffected muscles. This condition can also affect respiratory and bulbar muscles

Acute Transverse Myelitis (Transverse myelitis)

- Acute transverse myelitis is most commonly due to multiple sclerosis.

- It can occur in patients with vasculitis, systemic lupus erythematosus (SLE), antiphospholipid syndrome, other autoimmune disorders, mycoplasmal infections, Lyme disease, syphilis, tuberculosis (TB), COVID-19, or viral meningoencephalitis or in patients taking amphetamines, IV heroin, or antiparasitic or antifungal medications.
- Transverse myelitis occurs with optic neuritis in neuromyelitis optica (Devic disease), once considered a variant of multiple sclerosis but now considered a distinct disorder.
- The mechanism of transverse myelitis is often unknown, but some cases follow viral infection or vaccination, suggesting an autoimmune reaction.
- Inflammation tends to involve the spinal cord diffusely at one or more levels, affecting all spinal cord functions.

Clinical presentations-Symptoms and Signs

- Pain in the neck, back, or head.
- A bandlike tightness around the chest or abdomen, weakness, tingling, numbness of the feet and legs, and difficulty voiding develop over hours to a few days.
- Deficits may progress over several more days to a complete transverse sensorimotor myelopathy, causing paraplegia, loss of sensation below the lesion, urinary retention, and fecal incontinence.
- Occasionally, position and vibration sensation are spared, at least initially.
- Acute transverse myelitis occasionally recurs in patients with multiple sclerosis, SLE, or antiphospholipid syndrome.

Ancillary investigations

- MRI and cerebrospinal fluid (CSF) analysis
- Other tests to identify treatable causes
- Diagnosis of acute transverse myelitis is suggested by transverse sensorimotor myelopathy with segmental deficits.
- Guillain-Barre syndrome can be distinguished because it does not localize to a specific spinal segment.

- Diagnosis requires MRI and CSF analysis. MRI typically shows cord swelling if transverse myelitis is present and can help exclude other treatable causes of spinal cord dysfunction (eg, spinal cord compression).
- CSF usually contains monocytes, protein content is slightly increased, and IgG index is elevated (normal, = 0.85).
- A test for a marker for neuromyelitis optica IgG (NMO-IgG)
- Tests for treatable causes should include chest x-ray; PPD (purified protein derivative) for tuberculosis; serologic tests for mycoplasma, Lyme disease, COVID-19, and HIV; erythrocyte sedimentation rate (ESR) and/or C-reactive protein (CRP); antinuclear antibodies; and CSF and blood Venereal Disease Research Laboratory (VDRL) tests.
- History may suggest a medication or recreational drug as a cause.
- The differential diagnosis of acute transverse myelitis includes other transverse myelopathies due to nutritional deficiencies (eg, deficiency of vitamin B12, folate, zinc, or copper), vascular insufficiency, and intraspinal tumors.

Epilepsy Organic

History of Addiction

- Alcohol withdrawal seizures and seizures because of cocaine abuse are becoming more common.
- Patients frequently lie about their use of illicit drugs or alcohol. In young adults and teenagers, a urine drug screen should be done.

History of fever

- Seizures may be induced by high fevers in children who are otherwise normal and who never develop other neurologic problems, including epilepsy. However, febrile seizures occur only in a relatively small proportion of children
- Fever should make one think of meningitis or encephalitis, or if it has been extended over a longer period of time, a cerebral abscess.

- In children, one should consider the possibility of febrile convulsions.

Are there focal neurologic signs and papilledema

- These findings are more typical of a space-occupying lesion such as a cerebral tumor, cerebral abscess, or a subdural hematoma.

Epileptogenesis

- There are a variety of conditions that have an extremely high likelihood of resulting in a chronic seizure disorder. One of the best examples of this is severe, penetrating head trauma, which is associated with up to a 45% risk of subsequent epilepsy. The high propensity for severe traumatic brain injury to lead to epilepsy suggests that the injury results in a long-lasting pathologic change in the CNS that transforms a presumably normal neural network into one that is abnormally hyperexcitable. This process is known as epileptogenesis, and the specific changes that result in a lowered seizure threshold can be considered epileptogenic factors.
- Other processes associated with epileptogenesis include stroke, infections, and abnormalities of CNS development.

Other causes

- Cerebrovascular disease
- Metabolic disorders (uremia, hepatic failure, electrolyte abnormalities, hypoglycemia, hyperglycemia)
- Alzheimer's disease and other degenerative CNS diseases

Lumbago

Aetiology

Acute lumbar disc herniation is often precipitated by trauma, usually by lifting heavy weights while the spine is flexed.

Pathophysiology

The nucleus pulposus may bulge or rupture through the annulus fibroses



Giving rise to pressure on nerve endings in the spinal ligament



Changes in the vertebral joints or pressure on nerve roots



The altered mechanics of the lumbar spine



Result in loss of lumbar lordosis



And there may be spasm of the paraspinal musculature



Root pressure is suggested by limitation of flexion of the hip on the affected side



If the straight leg is raised (Lasegue's sign)



If the third or fourth lumbar roots are involved, Lasegue's sign may be negative



Pain in the back may be induced by hyperextension of the hip (femoral nerve stretch test)



The roots most frequently affected are S1, L-5 and L4

Clinical features

1. The onset may be sudden or gradual.
2. Alternatively, repeated episodes of low back pain may precede sciatica by months or years.
3. Constant aching pain is felt in the lumbar region and may radiate to the buttock, thigh, calf and foot.
4. Pain is exacerbated by coughing or straining but may be relieved by lying flat.

Physical signs in lumbar root compression

Disc level	Root loss	Sensory ness	Weak-loss	Reflex
L3/L4	L4	Inner calf	Inversion of foot	Knee
L4/L5	L5	Outer calf and dorsum of foot	Dorsiflexion of hallux/toes	
L5/S1	S1	Sole and lateral foot	Plantar flexion	Ankle

Investigations

1. Plain X-rays of the lumbar spine are of little value in the diagnosis of lumbar disc disease, although they may show other conditions such as malignant infiltration of a vertebral body.
2. CT, especially using spiral scanning techniques, can provide helpful images of the disc protrusion and/or narrowing of the exit foramina.
3. MRI is the investigation of choice if available, since soft tissues are well imaged.

Lumbar Spondylosis/ osteoarthritis lumbar spine

- Typically occurs in later life and primarily involves the cervical and lumbosacral spine.

Clinical features

- Patients often complain of back pain that increases with movement and is associated with stiffness.
- The relationship between clinical symptoms and radiologic findings is usually not straightforward.
- Osteophytes or combined disk-osteophytes may cause or contribute to central spinal canal stenosis, lateral recess stenosis, or neural foraminal narrowing

Investigations

- Lumbo-sacral x-ray (ap & lateral)
- Lumbo-sacral CT
- Lumbo-sacral MRI

Lumbar Disk Disease

1. Disk disease is most likely to occur at the L4-L5 or L5-S1 levels
2. The upper lumbar levels are involved occasionally.

Aetiology

3. The cause is often unknown
4. The risk is increased in overweight individuals.
5. Genetic factors may play a role in predisposing some patients to disk disease
6. This is a common cause of chronic or recurrent low back and leg pain.

7. Disk herniation is unusual prior to age 20 years and is rare in the fibrotic disks of the elderly.

Clinical features

- The pain may be located in the low back only or referred to a leg, buttock, or hip.
- A sneeze, cough, or trivial movement may cause the nucleus pulposus to prolapse, pushing the frayed and weakened annulus posteriorly.
- With severe disk disease, the nucleus may protrude through the annulus (herniation) or become extruded to lie as a free fragment in the spinal canal.

Investigations

- A lumbar spine MRI scan or CT-myelogram is necessary to establish the location and type of pathology.
- Spine MRIs yield exquisite views of intraspinal and adjacent soft tissue anatomy.
- Bony lesions of the lateral recess or intervertebral foramen are optimally visualized by CT-myelography.
- The diagnosis of nerve root injury is most secure when the history, examination, results of imaging studies, and the EMG are concordant.
- The correlation between CT and EMG for localization of nerve root injury is between 65 and 73%.
- Up to one-third of asymptomatic adults have a lumbar disk protrusion detected by CT or MRI scans.

Sciatica Syndrome

- Sciatica is a symptom and not a diagnosis.
- Sciatica describes leg pain that is localized in the distribution of one or more lumbosacral nerve roots, typically L4-S2, with or without neurological deficit.
- The leg pain from the lumbosacral segment is termed as sciatica.

Causes

- Degenerative arthritis and disc prolapse are the common causes.

- It is often caused by a herniated intervertebral disc, compressing the L5 or S1 ventral rami. The so-called piriformis syndrome, which may result from a variant or anomalous relationship between piriformis and the sciatic nerve, may also produce hip pain that radiates inferiorly along the course of the sciatic nerve.
- Broadly sciatica can either be because of inflammation of the sciatic nerve or because of compression of one of the extra constituting the sciatic nerve.

Inflammatory

- Sciatica neuritis
- Arachnoiditis

Nerve root compression

- Compression in the vertebral canal by disc, tumour, tuberculosis
- Compression in the intervertebral foramen due to root canal stenosis because of OA, spondylolisthesis, facet arthropathy or tumours
- Compression in the buttock or pelvis by abscess, tumour, haematoma.

Clinical features

- Sciatica is pain in the lower back and hip region, which radiates inferiorly along the posterior thigh to the leg.
- It means a pain radiating down the back of the thigh and calf.

Causes of Back or Neck Pain

Congenital/Developmental

- Spondylolysis and spondylolisthesis
- Kyphoscoliosis
- Spina bifida occulta
- Tethered spinal cord

Minor Trauma

- Strain or sprain
- Whiplash injury

Fractures

- Trauma-falls, motor vehicle accidents
- Atraumatic fractures-osteoporosis, neoplastic infiltration, exogenous steroids, osteomyelitis

Intervertebral Disk Herniation

- Degenerative
- Intervertebral foraminal narrowing
- Disk-osteophyte complex
- Internal disk disruption
- LSS with neurogenic claudication
- Uncovertebral joint disease
- Atlantoaxial joint disease (e.g., rheumatoid arthritis)
- Arthritis
- Spondylosis
- Facet or sacroiliac arthropathy
- Neoplasms-metastatic, hematologic, primary bone tumors
- Infection/Inflammation
- Vertebral osteomyelitis
- Spinal epidural abscess
- Septic disk (discitis)
- Meningitis
- Lumbar arachnoiditis
- Autoimmune [e.g., ankylosing spondylitis, reactive arthritis (formerly known as Reiter's syndrome)]
- Metabolic
- Osteoporosis-hyperparathyroidism, immobility
- Osteosclerosis (e.g., Paget's disease)
- Vascular
- Abdominal aortic aneurysm
- Vertebral artery dissection
- Other
- Referred pain from visceral disease
- Postural
- Psychiatric, malingering, chronic pain syndromes

Examination of the Back

- A physical examination that includes the abdomen and rectum is advisable.
- Back pain referred from visceral organs may be reproduced during palpation of the abdomen [pancreatitis, abdominal aortic aneurysm (AAA)] or percussion over the costovertebral angles (pyelonephritis).
- Inspection may reveal a lateral curvature of

the spine (scoliosis) or an asymmetry in the prominence of the paraspinal muscles, suggesting muscle spasm.

- Back pain of bony spine origin is often reproduced by palpation or percussion over the spinous process of the affected vertebrae.
- Forward bending is often limited by paraspinal muscle spasm; the latter may flatten the usual lumbar lordosis.
- Flexion at the hips is normal in patients with lumbar spine disease, but flexion of the lumbar spine is limited and sometimes painful.
- Lateral bending to the side opposite the injured spinal element may stretch the damaged tissues, worsen pain, and limit motion.
- Hyperextension of the spine (with the patient prone or standing) is limited when nerve root compression, facet joint pathology, or other bony spine disease is present.

Pain from Hip

- Pain from hip disease may mimic the pain of lumbar spine disease.

Patrick's sign

- Hip pain can be reproduced by internal and external rotation at the hip with the knee and hip in flexion (Patrick's sign)

Heel percussion sign

- Tapping the heel with the examiner's palm while the leg is extended (heel percussion sign).

Straight leg-raising maneuver

- With the patient supine, passive flexion of the extended leg at the hip stretches the L5 and S1 nerve roots and the sciatic nerve (straight leg-raising maneuver).
- Passive dorsiflexion of the foot during the maneuver adds to the stretch.
- While flexion to at least 80° is normally possible without causing pain, many patients normally report a tight, stretching sensation in the hamstring muscles unrelated to back pain.
- The *straight leg-raising (SLR)* test is positive if the maneuver reproduces the patient's usual back or limb pain.

- Eliciting the SLR sign in the sitting position can help determine if the finding is reproducible.
- The patient may describe pain in the low back, buttocks, posterior thigh, or lower leg, but the *key feature is reproduction of the patient's usual pain.*

Crossed SLR sign

- The *crossed SLR sign* is positive when flexion of one leg reproduces the usual pain in the opposite leg or buttocks.
- The crossed SLR sign is less sensitive but more specific for disk herniation than the SLR sign.
- The nerve or nerve root lesion is always on the side of the pain.

Reverse SLR sign

- The *reverse SLR sign* is elicited by standing the patient next to the examination table and passively extending each leg with the knee fully extended.
- This maneuver, which stretches the L2-L4 nerve roots, lumbosacral plexus, and femoral nerve, is considered positive if the patient's usual back or limb pain is reproduced.

Neurological examination findings

- It includes a search for focal weakness or muscle atrophy, focal reflex changes, diminished sensation in the legs, or signs of spinal cord injury.

Breakaway weakness

- It is defined as fluctuating strength during muscle testing.
- It may be due to pain or a combination of pain and underlying true weakness.
- Breakaway weakness without pain is almost always due to a lack of effort.
- In uncertain cases, electromyography (EMG) can determine whether or not true weakness due to nerve tissue injury is present.
- Findings with specific nerve lumbosacral nerve root lesions are discussed below.

Lumbosacral Radiculopathy : Neurologic Features

Examination Findings

Lumbosacral Nerve Roots	Reflex	Sensory	Motor	Pain Distribution	Sign
L2	-	Upper anterior thigh	Psoas (hip flexion)	Anterior thigh	Reverse straight leg-raising sign present
L3	-	Lower anterior thigh Anterior knee	Psoas (hip flexion) Quadriceps (knee extension) Thigh adduction	Anterior thigh, knee	Reverse straight leg-raising sign present
L4	Quadriceps (knee)	Medial calf	Quadriceps (knee extension) Thigh adduction Tibialis anterior (foot dorsiflexion)	Knee, medial calf Anterolateral thigh	Reverse straight leg-raising sign present
L5	-	Dorsal surface foot Lateral calf	Peronei (foot eversion) Tibialis anterior (foot dorsiflexion) Gluteus medius (hip abduction) Toe dorsiflexors	Lateral calf, dorsal foot, posterolateral thigh, buttocks	Straight leg-raising sign present
S1	Gastrocnemius/soleus (ankle)	Plantar surface-foot Lateral aspect-foot	Gastrocnemius/soleus (foot plantar flexion) Abductor hallucis (toe flexors) Gluteus maximus (hip extension)	Bottom foot, posterior calf, posterior thigh, buttocks	Straight leg-raising sign present

Brachial neuralgia /Brachial Neuritis

- Brachial neuritis (BN), also known as neuralgic amyotrophy or Parsonage-Turner syndrome, is a rare syndrome of unknown etiology affecting mainly the motor branches/fascicles of certain characteristic peripheral nerves in the arm.
- The term brachial may be a misnomer, as electrodiagnostic and radiologic evidence often points to a mononeuropathy or mononeuropathy multiplex rather than plexus involvement.
- Brachial neuritis usually is characterized by the acute onset of excruciating unilateral shoulder pain, followed by flaccid paralysis of shoulder and parascapular muscles several

days later. The syndrome can vary greatly in presentation and nerve involvement.

- Physical therapy for patients with brachial neuritis should be focused on the maintenance of full range of motion (ROM) in the shoulder and other affected joints.

Symptoms of brachial neuritis

- The onset of pain in brachial neuritis (BN) is often abrupt and may follow recent illness, surgery, immunization, or even trauma. Up to two thirds of cases begin during the night time.
- The pain usually is localized to the right shoulder region, but it may be bilateral in 10-30% of cases. The pain's intensity is very high (9+/10) and is maximal at onset. Usually, the pain is described as sharp or throbbing in nature.

- The pain usually is constant, but it is exacerbated by movements of the shoulder. Movements of the neck, coughing, and/or sneezing usually do not worsen the pain.
- Intense pain can last from a few hours to several weeks and requires opiate analgesia.
- Low-grade pain may persist for up to a year. As pain subsides, weakness becomes apparent.
- Numbness may occur, depending on the particular nerves affected, and usually is found in the nerve distribution corresponding to maximal muscle weakness. However, numbness is rarely a prominent complaint.

Ancillary investigations

- Laboratory studies are indicated in brachial neuritis only if systemic disease is suspected on clinical grounds.
- The complete blood count (CBC) and erythrocyte sedimentation rate (ESR) can be used as nonspecific indicators of systemic disease.
- Antinuclear antibody (ANA) values can be used as a marker for connective-tissue disease.
- Human immunodeficiency virus (HIV) serology can also be performed.
- Magnetic resonance imaging (MRI) or computed tomography (CT) myelogram scanning should be considered initially to rule out cervical radiculopathy (particularly C5/C6).
- MRI of the brachial plexus can help to rule out carcinomatous or granulomatous infiltration, if clinically indicated.
- A shoulder radiograph may be indicated to rule out specific shoulder pathologies.
- Electrodiagnosis should be considered initially to confirm neuropathic diagnosis and to rule out various other conditions (eg, radiculopathy, neuropathy, amyotrophic lateral sclerosis).

Cervical spondylosis

Cervical spondylosis

1. Cervical spondylosis is the term given to the occurrence of osteoarthritis in the cervical spine.

2. It is characterised by degeneration of the intervertebral discs and osteophyte formation.
3. This extremely common and radiological changes of cervical spondylosis are very frequently found in apparently healthy individuals above the age of 50.
4. Whilst cervical spondylosis is often asymptomatic, it may be associated with neurological dysfunction.
5. The C5/6, C6/7 and C4/5 vertebral levels and C6, C7 and C5 roots, respectively, are most commonly affected.

Cervical spondylotic radiculopathy

Compression of a nerve root occurs when disc prolapses laterally, which may develop acutely, or compression may occur more gradually due to osteophytic encroachment of the intervertebral foramina.

Clinical features

1. The patient complains of pain in the neck that movement in the distribution of the affected nerve root.
2. The neck is held rigidly and neck movements may exacerbate pain.
3. Paraesthesia and sensory loss may be found in the affected segment
4. There may be lower motor neuron signs, including weakness, wasting and reflex impairment

Physical signs in cervical root compression

Root	Muscle weakness	Sensory loss	Reflex loss
C5	Biceps, deltoid, spinati	Upper lateral arm	Biceps
C6	Brachioradialis	Lower lateral arm, thumb, index finger	Supinator
C7	Triceps, finger and wrist extensors	Middle finger	Triceps

Investigations

1. Plain X-rays, including lateral and oblique views, should be obtained to confirm, the

presence of degenerative changes and to exclude other conditions, including destructive lesions.

2. If surgery is contemplated, MRI is required.
3. Electrophysiological studies rarely add to the clinical examination, but may be necessary if there is doubt about the differential diagnosis between root and peripheral nerve lesions.

Lumbar spondylosis (Spondylosis deformans, lumbar osteoarthritis)

- The term spondylolysis is derived from the Greek words spondylos, meaning vertebra, and lysis, meaning break or defect.
- It describes bony overgrowths (osteophytes, bone spurs), predominantly those at the anterior, lateral, and, less commonly, posterior aspects of the superior and inferior margins of vertebral centra(bodies).
- Lumbar spondylosis occurs as a result of new bone formation in areas where the annular ligament is stressed. The margins of vertebral bodies are normally smooth. Growth of new bone projecting horizontally at these margins identifies osteophytes.
- Most osteophytes are anterior or lateral in projection. Posterior vertebral osteophytes are less common and only rarely impinge upon the spinal cord or nerve roots. This dynamic process increases with, and is perhaps an inevitable concomitant, of age. Lumbar osteophytes have long been thought to cause back pain because of their frequency and size. This has led to many studies of the distribution of vertebral osteophytes, not all of which are pertinent. The frequency of signs or symptoms in individuals with osteophytes is no greater than that in individuals without osteophytes.
- Lumbosacral spondylolysis (lumbar spondylolysis) is a unilateral or bilateral defect of the pars interarticularis that affects one or more of the lumbar vertebrae
- *Spondylosis*, or osteoarthritic spine disease, typically occurs in later life and primarily involves the cervical and lumbosacral spine.
- Patients often complain of back pain that increases with movement and is associated

with stiffness.

- The relationship between clinical symptoms and radiologic findings is usually not straightforward.
- Pain may be prominent when x-ray, CT, or MRI findings are minimal, and prominent degenerative spine disease can be seen in asymptomatic patients.
- Osteophytes or combined disk-osteophytes may cause or contribute to central spinal canal stenosis, lateral recess stenosis, or neural foraminal narrowing.

Epidemiology

Spondylolysis occurs in 6-10% of the general population and has been found to be as high as 25-60% in athletes. It is especially common in young athletes younger than 18 years who participate in sports that involve twisting or backward bending motions of the spine. This injury also runs in some families, suggesting that there may be a hereditary component

Clinical presentation

- Among the general population, most cases of lumbosacral spondylolysis (lumbar spondylolysis) are clinically inapparent, and the condition is symptomatic in only 10% of patients.
- Many athletes with lumbosacral spondylolysis (lumbar spondylolysis) are likewise asymptomatic.
- Lumbosacral spondylolysis (lumbar spondylolysis) typically occurs in young people, with a mean age at diagnosis in athletes of about 15-16 years.
- Patients with lumbosacral spondylolysis (lumbar spondylolysis) generally report low back pain aggravated by activity, particularly with hyperextension maneuvers such as in gymnastics.
- In general, patients present without a history of neurologic symptoms.

Physical Examination

The physical examination of the patient with lumbosacral spondylolysis (lumbar spondylolysis) frequently yields the following minimal findings:

- No tenderness to palpation is noted, but some discomfort can be elicited with deep percussion over the midline of the lumbar area.
- Range of motion is full. Because of the previously increased range of motion of dancers and gymnasts, they appear to have normal flexibility.
- Athletes suspected of having an injury must have their flexibility compared with that of their preinjury state.
- Forward flexion does not increase symptoms.
- Hyperextension mimicking the sporting movement generally elicits pain.
- Sciatica can occur but is rare.
- Physical findings that may also be present include antalgic or normal gait, tight lumbar musculature and hamstrings, hyperlordosis, and buttock or thigh pain.
- If a unilateral defect is present, the 1-leg hyperextension test elicits pain on the involved side. This test is performed by the patient bearing weight on one leg, with both the hip and knee of the other extremity flexed while hyperextending the lumbar spine.
- The maneuver is performed on both sides, and asymmetric low back pain indicates unilateral disease. Bilateral disease may show symmetric or asymmetric pain with this maneuver.
- The neurovascular examination findings are normal. The rest of the general low back pain examination must be performed, taking note of all dermatomes, myotomes, and reflexes.

Ancillary investigations

Imaging Studies

- Routine thoracolumbar radiography in an athlete suspected of having a lumbosacral spondylolysis (lumbar spondylolysis) should include anteroposterior (AP), lateral, and oblique views.
- Flexion and extension lateral radiographs can be added if spondylolisthesis is present or stability is in question.

- AP radiographic findings are usually normal in spondylolysis.
- The oblique views are particularly screened for the "Scotty dog" lesion in the pars interarticularis. The pars defect is represented by the collar on the Scotty dog. Keep in mind that the oblique view must be of good quality and read by someone experienced in making the diagnosis of lumbosacral spondylolysis (lumbar spondylolysis).
- An athlete who is symptomatic with hyperextension and whose plain radiography results are normal requires additional investigations. A 99-m technetium (99m Tc) ethylene diphosphonate bone scintigraphy identifies pars interarticularis stress fractures that were missed on oblique plain radiography. Those patients who have had recent trauma or strenuous activity and who are symptomatic show increased activity on the spondylolytic area. On the other hand, those with chronic low back pain show normal scans.
- Technetium bone scanning has its limitations. When compared to single-photon emission computed tomography (SPECT) scanning, the SPECT scanning was found to be more sensitive and better at localization of abnormalities in those athletes with radiographically obvious lumbosacral spondylolysis/spondylolisthesis.
- Bone scanning has a false-positive rate of 15% when using computed tomography (CT) scanning as the standard to establish the diagnosis of lumbosacral spondylolysis (lumbar spondylolysis). Therefore, SPECT scanning is needed to identify the more subtle stress reactions.

Sagittally reconstructed computed tomography scan

- Sagittally reconstructed computed tomography scan of the lumbar spine shows a defect of the pars interarticularis on the left at L5.
- CT scanning also provides information regarding other features of spondylolysis, such as facet joint changes, spondylolisthesis, and disc herniations, and it also helps prognosticate the potential for a lesion to heal.

- MRI is inferior to CT scanning for direct visualization of defects of the pars interarticularis. However, MRI has utility to rule out disc herniations.

Long TR (T2-weighted) fat suppressed sagittal magnetic resonance

- Long TR (T2-weighted) fat suppressed sagittal magnetic resonance image shows increased signal in the pars interarticularis on the left at L5
- This is an acute stress reaction.

Sagittal short TR (T1-weighted) magnetic resonance

Sagittal short TR (T1-weighted) magnetic resonance image shows decreased signal in the pars interarticularis on the left at L5

MCQs

- Parkinson's disease (PD) causes damage to the—— and is associated with impairments in habit learning but spares ——
 - Hippocampal episodic Memory
 - Bilateral anterior temporal dysfunction Long-Term Memory
 - Basal ganglia declarative memory
 - Basal ganglia Nondeclarative Memory
- Burning sensation in one half of the body (Hyperalgesia) is seen in
 - Thalamic stroke
 - Lateral medullary syndrome
 - Cerebellar lesion
 - Meningitis
- Which cranial nerve is affected in Bell's palsy?
 - Trigeminal nerve (Cranial nerve V)
 - Facial nerve (Cranial nerve VII)
 - Vagus nerve (Cranial nerve X)
 - All of the above
- The upper motor neuron impairment produces the following change of muscles tone:
 - Flaccidity
 - Spasticity
 - "Cog wheel" rigidity
 - Myoclonia
- The muscular wasting (hypotrophy) usually develops with disease in:
 - Upper motor neuron
 - Lower motor neuron
 - Cerebellar
 - Caudate
- The ability to walk along a straight line is most often impaired with:
 - Cerebellar dysfunction
 - Parietal lobe damage
 - Temporal lobe damage
 - Ocular motor disturbances
- Parkinsonism includes combination of the following:
 - Tremor, bradykinesia & muscles rigidity
 - Paresis, anesthesia & muscles spasticity
 - Chorea & muscles hypotonia
 - Tremor, ataxia & muscles hypotonia
- Meningeal sign is the following:
 - Babinsky
 - Kernig
 - Lasseg
 - Romberg
- Elevated IgG index level in CSF is seen in which among the following disease
 - Acute transverse myelitis
 - Syringomyelia
 - Conus medullaris syndrome
 - Spinal cord abscess
- Parsonage-Turner syndrome is also known as
 - Brachial neuritis
 - Amyotrophic Lateral Sclerosis
 - Multiple sclerosis
 - Parkinson's disease

Answer

1.	c	2.	a	3.	b
4.	b	5.	b	6.	a
7.	a	8.	b	9.	a
10.	a				